

MD MUSHFIQUR RAHMAN

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Jackson, Mississippi 39202

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EDUCATION

University of Mississippi Medical Center 2024 – Present
PhD, Department of Data Science Jackson, Mississippi
Third-year student (currently enrolled). Research focus: Functional Data Analysis, Functional Regression.

University of Mississippi Medical Center 2024 – 2026
MS, Biostatistics and Data Science Jackson, Mississippi

University of Dhaka 2018 – 2023
BS, Department of Statistics Dhaka, Bangladesh

EXPERIENCE

MIND Center, University of Mississippi Medical Center 2024 – Present
Student Biostatistician (Part-Time)

- Provide statistical support for neuroscience and aging research, with a focus on gait speed and cognitive status.
- Apply functional regression and longitudinal methods to complex biomedical datasets.
- Collaborated on the project “Associations of Alzheimer’s Disease and Cerebrovascular Disease Biomarkers with Cognitive, Mobility, and Dual Impairment in the ARIC Study” (2025), presented by MSTAR student Meredith Boles.
- Collaborate with clinical researchers on study design, analysis, and scientific reporting.

icddr,b October 2023 – June 2024
Data Management Assistant

- Organized and managed large datasets ensuring accurate and efficient data processing.
- Utilized STATA, SPSS, and R for storage, analysis, and visualization of clinical data.
- Assisted in several interim analyses of clinical trials to assess the efficacy of multiple new drugs.
- Analyzed Diarrheal Disease Surveillance System (DDSS) data at the Dhaka Hospital.
- Participated in a Phase III double-blinded randomized controlled trial using a probiotic to monitor weight gain for moderately and severely stunted children (2 months to 1 year of age).
- Built a Shiny app to showcase and analyze various aspects of the study.

Save the Children in Bangladesh May 2023 – July 2023
Research Assistant

- Performed analysis to measure maternal and newborn care indicators (ANC visits, essential newborn care, place and mode of delivery, postpartum family planning).
- Conducted multivariate analysis on factors contributing to ANC utilization, mode of delivery, and newborn care practices.

- Assessed intervention effectiveness using the LiST (The Lives Saved Tool).
- Managed survey data and created a data dictionary.

Statistical Research Consultancy Bangladesh (SRCBD)
Trainee Data Analyst (Internship)

November 2020 – July 2021

- Aided in course curriculum design for R programming.
- Performed data visualization using `ggplot2` and hypothesis testing in R.

RESEARCH & PUBLICATIONS

- [1] Hassan, Md. Mehedi and Mamun Billah, Md. Al and **Rahman, Md. Mushfiqur** and Zaman, Sadika and Hasan Shakil, Md. Mehadi and Angon, Jarif Huda. **Early Predictive Analytics in Healthcare for Diabetes Prediction Using Machine Learning Approach**. *2021 12th International Conference on Computing Communication and Networking Technologies (ICCCNT)*, 2021, pp. 01–05. DOI: [10.1109/ICCCNT51525.2021.9579799](https://doi.org/10.1109/ICCCNT51525.2021.9579799).
- [2] Hassan, Md. Mehedi and Hassan, Md. Mahedi and Akter, Laboni and **Rahman, Md. Mushfiqur** and Zaman, Sadika and Khan, Md. Hasib and Jahan, Nusrat and Smrity, Raisun Nasa and Farhana, Jerin and Raihan, M. and Mollick, Swarnali. **Efficient Prediction of Water Quality Index (WQI) Using Machine Learning Algorithms**. *Human-Centric Intelligent Systems*, vol. 1, pp. 86–97, 2021. DOI: [10.2991/hcis.k.211203.001](https://doi.org/10.2991/hcis.k.211203.001).
- [3] Hassan, Md. Mehedi and Raihan, M. and Khan, Md. Hasanul Banna and Dhali, Tutul and **Rahman, Md. Mushfiqur** and Sneha, Zarmina Huda. **A Hybrid Machine Learning Approach to Predict the Risk of Having Stroke**. *ACM*, 2022, pp. 460–465. DOI: [10.1145/3542954.3543020](https://doi.org/10.1145/3542954.3543020).
- [4] Hassan, Md. Mehedi and Zaman, Sadika and **Rahman, Md. Mushfiqur** and Bairagi, Anupam Kumar and El-Shafai, Walid and Rathore, Rajkumar Singh and Gupta, Deepak. **Efficient Prediction of Coronary Artery Disease Using Machine Learning Algorithms with Feature Selection Techniques**. *Computers and Electrical Engineering*, vol. 115, 109130, 2024. DOI: [10.1016/j.compeleceng.2024.109130](https://doi.org/10.1016/j.compeleceng.2024.109130).
- [5] Md Fuad Al Fidah and Md Ridwan Islam and **Rahman, Md. Mushfiqur** and Mohammad Jobayer Chisti and ASG Faruque and Tahmeed Ahmed and Sharika Nuzhat. **Community-Level Expenses for Under-5 Diarrhoeal Cases Prior to Reporting at Dhaka Hospital: A Cross-Sectional Analysis in Bangladesh**. *Clinical Epidemiology and Global Health*, vol. 31, 101846, 2025. DOI: [10.1016/j.cegh.2024.101846](https://doi.org/10.1016/j.cegh.2024.101846).
- [6] Bashar, Syed Jayedul and Islam, Md Ridwan and Nuzhat, Sharika and Amin, Rukaeya and **Rahman, Md. Mushfiqur** et al. **Antibiotic Use Prior to Attending a Large Diarrheal Disease Hospital Among Preschool Children Suffering from Bloody or Non-Bloody Diarrhea: A Cross-Sectional Study Conducted in Bangladesh**. *PLOS ONE*, 19(11): e0314325, 2024. DOI: [10.1371/journal.pone.0314325](https://doi.org/10.1371/journal.pone.0314325).
- [7] Nuzhat, Sharika and Islam, Md Ridwan and Fidah, Md Fuad Al and Islam, Shoeb Bin and **Rahman, Md Mushfiqur** and Paul, Sneha and Nesa, Mahfuz-Un and Chowdhury, Devjani and Nabi, Shayontan Fardin and Awn, Awni Mokhtar Sallam Ali and Khan, Liaquat Ali and Kawser, Chowdhury Ali and Chisti, Mohammad Jobayer and Ahmed, Tahmeed. **Maternal Knowledge, Attitude and Practice Regarding Commercial Oral Rehydration Salt Solution: Experience from a Diarrhoeal Disease Hospital in Bangladesh**. *BMJ Paediatrics Open*, 9(1): e003299, 2025. DOI: [10.1136/bmjpo-2024-003299](https://doi.org/10.1136/bmjpo-2024-003299).

- [8] Nuzhat, Sharika and **Rahman, Md. Mushfiqur** et al. **Trends in Preventive Practices Against Diarrhoeal Disease Among Under-Five Children: Experience from the Largest Diarrhoeal Disease Hospital in Bangladesh.** *BMJ Paediatrics Open*, 9(1): e003259, 2025. [PMC11979600](#).
- [9] Rahman, Amrina and **Rahman, Md. Mushfiqur** and Yasmin, Farhana. **Revolutionizing Healthcare through Federated Learning.** In: *A Secure and Trusted Approach for Federated Learning*, Chapter 1. Taylor & Francis / CRC Press, 2024. [Publisher Link](#).

PRESENTATIONS & CONFERENCES

- **Quantifying Neurocognitive Effects of Sleep Stage Architecture in OSA: A Variable Domain Functional Regression Approach.** *ASA Alabama–Mississippi Chapter.* 2026
2nd Place — Best Presentation Award.
- **Annual Meeting.** ASA Alabama–Mississippi Chapter. 2025
- **70th Annual Meeting.** Mississippi Academy of Science. 2025
- **69th Annual Meeting.** Mississippi Academy of Science. 2024

PROJECTS & REPORTS

- **Associations of Alzheimer’s Disease and Cerebrovascular Disease Biomarkers with Cognitive, Mobility, and Dual Impairment in the ARIC Study.** MIND Center, UMMC (collaborative project with MSTAR student Meredith Boles, presented 2025). 2025
- **Early Childhood Development: Variation within Age at Marriage and Birth of Children among Women in Bangladesh.** BS Final Year Project, University of Dhaka. 2022
- **Socio-Demographic Factors Affecting Maternal Health Care in Bangladesh.** 2020

DATA ANALYTICS SKILLS

Programming Languages	R, STATA, SPSS, SAS, MATLAB, Python, PowerBI
R Packages	ggplot2, dplyr, tidyr, caret, tidymodels, shiny, Quarto, survival, maptools
Software & Tools	Word, L ^A T _E X, Excel, PowerPoint

ACHIEVEMENTS

- 2nd Place — Best Presentation Award** 2026
SOPH Research Day, University of Mississippi Medical Center.
- Government Honours Scholarship** 2024
Directorate of Secondary and Higher Education, People’s Republic of Bangladesh.
- Best Paper Award** 2022
Human-Centric Intelligent System, Springer.
- Winner, Data Analysis Competition** 2020
QMH Statistics Club, Department of Statistics, University of Dhaka.